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Lean Combustion with Near-Critical Swirling Flows

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Abstract

The dynamics of a lean-premixed combustion state in a straight circular tube with an underlying low Mach number near-critical axisymmetric swirling flow is studied. A small-disturbance approach is developed to carefully explore the nonlinear interaction between the flow swirl and the heat transfer through the combustion. The analysis shows that in the leading order the reaction problem can be decoupled from the flow problem. However, the effect of the heat addition on the flow behavior is important and has a nonlinear influence. It is found that a small but finite heat addition to the flow changes the transcritical bifurcation of solutions of the Euler equations for a cold (non-reactive) flow into two branches of solutions of the reactive flow equations. These branches fold at limit swirl levels near the critical swirl resulting in a finite gap of swirl that separates the two branches. This suggests that no near-columnar equilibrium state can exist for an incoming flow with swirl close to the critical level and the flow must develop large disturbances and a vortex breakdown process in this swirl range. Also, the critical swirl for a reactive flow is less than that for a cold flow. Beyond this range of swirl, two equilibrium states may exist under the same inlet/outlet conditions and combustion state. One which describes a near-columnar state, and another which shows the appearance of a large-amplitude solitary wave. When the heat addition is increased, this special behavior uniformly changes into a branch of equilibrium states with no fold. It shows the gradual appearance of large-amplitude waves in the flow as the swirl level is increased. The relevance of the results to combustion states with vortex breakdown is also discussed.

I. Introduction

Combustion is susceptible to a variety of destabilizing influences (Williams,[1] Kapila,[2] Gutmark et al.,[3,4] McManus et al.,[5] Paschereit et al.[6]). These can either be intrinsic to the mode of combustion, or arise as a result of interaction between the combustion process and its flow environment. A manifestation of intrinsic instability is the spontaneous development of cellular structures in flames. Instabilities induced by the coupling between combustion and the underlying flow are exhibited by reactive shear layers and jets, while acoustic instabilities in propulsion devices, and flame acceleration prior to possible transition to detonation in closed vessels, are examples where confining boundaries and obstacles in the path of flames play crucial roles. In technological applications, instabilities can have an adverse effect (e.g., damage or flame extinction), a favorable effect (e.g., enhanced burning) or be of no consequence.

Instabilities in which combustion interacts with the underlying flow, the boundaries, or with processes occurring in other parts of the system can involve a broad range of physical mechanisms, including acoustic oscillations, vortex-shedding, and deformation of the flame zone. One may not have a simple, basic, steady configuration that can be subjected to the standard techniques of stability analysis. The problem is particularly challenging when the underlying inert flow is itself susceptible

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to instabilities, such as a swirling flow. It is not at all clear then what the role played by combustion is, and how the process of combustion itself is affected as a result of the intrinsic instability of the underlying flow.

Swirling flows are of practical importance in enhancing the efficiency of certain lean-premixed combustion systems. A promising technique is to induce axisymmetric vortex breakdown in the incoming flow of a mixture of air and fuel (Lefebvre [7]). This technique is used in various industrial combustors such as those designed by Pratt and Whitney, Ellison, and ABB (see, for example, McVey et al., [8] Snyder et al., [9] Stephens et al., [10–12] Wang and Yang, [13] and Paschereit et al. [14–15]). The technique is based on the idea that when the swirl ratio of the flow is above a certain critical level, a large and nearly stagnant axisymmetric separation region appears near the inlet of the combustion chamber. In the case of a premixed flux of air and fuel, the burned particles are trapped in the breakdown zone where they create a region of high temperature. Heat transfer from this hot region to the surrounding swirling flow apparently helps to stabilize the flame and burn more of the reactants, thereby improving combustion effectiveness (Sivasegaram and Whitelaw [16]). In the cases where liquid fuel is injected into a swirling flow of air, the hot breakdown zone serves as a noninvasive flame holder where fuel particles are trapped for longer residence time and mixing between air and fuel is improved.

However, a major problem in inducing the axisymmetric vortex breakdown in combustion states is the appearance of flow instabilities inside the large separation zone and in its wake (see, for example, Syred et al., [17] Gupta et al., [18] Gupta et al. [19]) At certain critical situations, the breakdown zone itself may become unstable and disappear because of changes in the distribution of swirl of the incoming flow, variations in the temperature field, or large perturbations in the back pressure at the downstream of the combustor. In such situations, the fluid mechanical instabilities also induce flame instabilities and affect the combustor performance. Gupta et al. [18,19] discussed different modes of instabilities generated in combustion with swirl.

It is therefore clear that a detailed study of the dynamical behavior of combustion systems with swirl is needed. A theoretical study of such systems is crucial for a better physical understanding and mathematical modeling of the mechanisms underlie the instabilities and the critical conditions for their appearance. It may also provide new strategies to improve the control of the combustion system.

As a topic in fluid mechanics, vortex breakdown boasts an extensive literature, including reviews by Hall, [20] Leibovich, [21] Escudier, [22] Sarpkaya, [23] Althaus et al., [24] and Rusak. [25] However, a consistent description of the phenomenon has proved elusive. Several possible explanations have been advanced, each clarifying some aspects of the problem. The relationship between various theoretical and numerical solutions has never been fully clarified. Until recently, no clear criteria for the occurrence, stability and dynamics of vortex breakdown states were given.

Recently, Rusak and co-authors have developed a novel theoretical framework for predicting the axisymmetric vortex breakdown process (see Wang and Rusak, [26–29] Rusak and Wang, [30] Rusak et al., [31–33] and Rusak [34,35]). This approach, employing the Euler and Navier-Stokes formulations, examines the dynamics of axisymmetric swirling flows in an axisymmetric chamber of finite length. The results, established through a rigorous nonlinear global analysis, provide a fundamental and nearly complete mathematical description of the dynamics and stability of inert axisymmetric swirling flows. The results show a good correlation with numerical computations and experimental studies (Malkiel et al. [36] and Rusak et al. [32,33]). It is also found that the theory naturally unifies the major previous theoretical and numerical approaches to the ax-

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isymmetric vortex breakdown process, fills the void between the various theories and thus provides a global understanding of the problem (Rusak[25]).

The approach of Wang and Rusak[28] shows that in the inviscid flow limit, there exist two critical swirl ratios of the incoming flow to the pipe, ω_0 and ω_1, where $\omega_0 < \omega_1$. Columnar flows with a swirl ratio less than the threshold level, ω_0, are unconditinally stable to any axisymmetric disturbance. In the range $\omega_0 < \omega < \omega_1$ the flow may evolve into one of two steady states, depending on the size of the initial disturbances. When disturbances are sufficiently small they decay in time and the flow returns to be columnar but, when disturbances are large, they grow and evolve into a large and steady, semi-infinite stagnation zone, similar to the breakdown states found in very high Reynolds number flows. When $\omega > \omega_1$, any initial disturbance grows and evolves into a breakdown zone.

These special stability characteristics are related to the upstream propagation of both small- and large-amplitude disturbances and their interaction with the flow conditions downstream of the vortex generator. The disturbances tend to propagate upstream with a speed that increases with ω. When $\omega < \omega_1$, small disturbances are convected by the axial flow out of the pipe and the columnar flow is, therefore, stable. However, when $\omega > \omega_1$, small disturbances tend to move upstream. Since the flow out of the vortex generator at steady operation is fixed, the disturbances cannot move through it. They are trapped, grow, and stabilize as a large and steady stagnation zone. These large-amplitude zones also tend to grow and move upstream as ω changes. However, when $\omega < \omega_0$, they are convected by the axial flow out of the pipe and the flow returns to be columnar.

The theory also presents for the first time a consistent explanation of the physical mechanism leading to axisymmetric vortex breakdown in a pipe as well as the conditions for its occurrence. The results show that this phenomenon is a necessary evolution from an initial columnar concentrated vortex flow, to another relatively stable lower-energy equilibrium state describing a swirling flow around a large breakdown zone. This evolution is the result of the interaction between waves propagating upstream, and the incoming flow, which leads to an absolute loss of stability of the base columnar state when the swirl ratio of the incoming flow is near or above a critical level. It should be noted here that the theory also shows that the fundamental characteristics which lead to vortex instability and breakdown in high Reynolds number vortex flows in a pipe, as well as the conditions where the breakdown state itself becomes unstable (the critical conditions), may be calculated from a single (reduced-order) nonlinear ordinary differential equation.

To the best of our knowledge, fundamental studies on combustion with swirling flows are quite limited to experimental and numerical simulations (Sivasegaram and Whitelaw,[16] Wang and Yang[13], Paschereit et al.[14–15] and Gupta et al.[18,19]). Specifically, there is no consistent analysis of the interaction between the combustion and the vortex breakdown processes. The effect of heat release on the stability and dynamics of swirling flows is unclear.

In the present paper we develop a first step toward an understanding of the dynamics of combustion states with swirl. We study the dynamics of a lean-premixed combustion state in a straight pipe with an underlying near-critical (with swirl level around ω_1) and near-columnar axisymmetric swirling flow. To achieve this goal, an asymptotic analysis is proper as well as necessary. The analysis is based on the low Mach number theory for reactive flows. A small-disturbance approach is developed to carefully explore the nonlinear interaction between the flow swirl and the heat transfer through combustion. The analysis reveals the possible states that may develop at various swirl levels and combustion parame-

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ters. The relevance of the results to combustion states with vortex breakdown is also discussed.

II. Mathematical model

A combustion state with swirling flow in a straight circular tube is considered. The tube radius is $\bar{r}_0$ and length is $\bar{x}_0$. The equations governing the steady, axisymmetric, compressible, and inviscid flow of a reactive premixed fluid are, in dimensional form,

$$\bar{p} = \bar{\rho} R \bar{T}, \tag{1}$$

$$(\bar{\rho}\bar{u})_{\bar{r}} + \frac{\bar{\rho}\bar{u}}{\bar{r}} + (\bar{\rho}\bar{w})_{\bar{x}} = 0, \tag{2}$$

$$\bar{\rho}\left(\bar{u}\bar{u}_{\bar{r}} + \bar{w}\bar{u}_{\bar{x}} - \frac{\bar{v}^2}{\bar{r}}\right) = -\bar{p}_{\bar{r}}, \tag{3}$$

$$\bar{u}\bar{v}_{\bar{r}} + \bar{w}\bar{v}_{\bar{x}} + \frac{\bar{u}\bar{v}}{\bar{r}} = 0, \tag{4}$$

$$\bar{\rho}\left(\bar{u}\bar{w}_{\bar{r}} + \bar{w}\bar{w}_{\bar{x}}\right) = -\bar{p}_{\bar{x}}, \tag{5}$$

$$\bar{\rho}\left(\bar{u}\bar{Y}_{\bar{r}} + \bar{w}\bar{Y}_{\bar{x}}\right) = D\left(\frac{1}{\bar{r}}\left(\bar{r}\bar{Y}_{\bar{r}}\right)_{\bar{r}} + \bar{Y}_{\bar{x}\bar{x}}\right) - W, \tag{6}$$

$$\bar{\rho}C_p\left(\bar{u}\bar{T}_{\bar{r}} + \bar{w}\bar{T}_{\bar{x}}\right) - \left(\bar{u}\bar{p}_{\bar{r}} + \bar{w}\bar{p}_{\bar{x}}\right) = \lambda\left(\frac{1}{\bar{r}}\left(\bar{r}\bar{T}_{\bar{r}}\right)_{\bar{r}} + \bar{T}_{\bar{x}\bar{x}}\right) + BW, \tag{7}$$

$$W = A\bar{\rho}\bar{Y}\exp(-\bar{E}/R\bar{T}). \tag{8}$$

Here $\bar{p}$ is the pressure, $\bar{\rho}$ the density and $\bar{T}$ the temperature of the fluid, while $\bar{u}$, $\bar{v}$, and $\bar{w}$ are, respectively, the radial, circumferential, and axial components of the velocity. $\bar{Y}$ denotes the mass fraction of the reactant in the combustible gas. Thermophysical properties of the reacting fluid include the specific gas constant R, the diffusion coefficient of the reactant D, the specific heat at constant pressure C_p, the thermal conductvity λ and the specific heat release B. These parameters are all assumed to be constants. A one-step first-order Arrhenius reaction is postulated, with $\bar{A}$ the reaction rate pre-exponential factor and $\bar{E}$ the activation energy.

We render the equations dimensionless by choosing the following scales: $\bar{r}_0$ the reference length, $\bar{T}_0$ the reference temperature, $\bar{\rho}_0$ the reference density, $\bar{p}_0 = \bar{\rho}_0 R\bar{T}_0$ the reference pressure, and $\bar{U}_0$ the reference speed. We select the reference scales as those characterizing the inlet stream. In particular, the incoming mixture is taken to be at a nominally uniform thermodynamic state, $\bar{T}_0$ and $\bar{\rho}_0$. The axial velocity, allowed to vary with radius, is taken to have $\bar{U}_0$ as its maximum. The equations, in dimensionless form, become

$$p = \rho T, \tag{9}$$

$$(\rho u)_r + \frac{\rho u}{r} + (\rho w)_x = 0, \tag{10}$$

$$\rho\left(uu_r + wu_x - \frac{v^2}{r}\right) = -\frac{1}{\gamma M_0^2}p_r, \tag{11}$$

$$uv_r + wv_x + \frac{uv}{r} = 0, \tag{12}$$

$$\rho\left(uw_r + ww_x\right) = -\frac{1}{\gamma M_0^2}p_x, \tag{13}$$

$$\rho\left(uY_r + wY_x\right) = \frac{1}{P_e}\left(\frac{1}{r}\left(rY_r\right)_r + Y_{xx}\right) - W, \tag{14}$$

$$\rho\left(uT_r + wT_x\right) - \frac{\gamma-1}{\gamma}\left(up_r + wp_x\right) = \frac{\mathcal{L}}{P_e}\left(\frac{1}{r}\left(rT_r\right)_r + T_{xx}\right) + \beta W, \tag{15}$$

$$W = \mathcal{A}\rho Y\exp(-\theta/T). \tag{16}$$

The dimensionless parameters appearing in these equations are $M_0 = \sqrt{\frac{\bar{U}_0}{\gamma\bar{p}_0/\bar{\rho}_0}}$ the reference Mach number, $P_e = \frac{\bar{\rho}_0\bar{U}_0\bar{r}_0}{D}$ the Peclet number, $\mathcal{L} = \frac{\lambda}{C_p D}$ the Lewis number, $\beta = \frac{B}{C_p\bar{T}_0}$ the heat-release parameter, $\theta = \frac{\bar{E}}{R\bar{T}_0}$ the activation temperature, and $\mathcal{A} = \frac{\bar{A}\bar{r}_0}{\bar{U}_0}$, the reaction-rate frequency factor.

We consider the following boundary conditions. Along the tube axis $r = 0$ the symmetry conditions

$$u(x,0) = v(x,0) = T_r(x,0) = Y_r(x,0) = 0 \tag{17}$$

apply for every $0 \le x \le x_0$. At the tube wall $r = 1$ the normal velocity vanishes as do the

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fluxes of heat and reactant, i.e.,

$$u(x,1) = T_r(x,1) = Y_r(x,1) = 0 \qquad (18)$$

for every $0 \le x \le x_0$. At the tube inlet $x = 0$ we consider a steady flow state that is continuously generated by a swirl generator ahead of the tube. We prescribe general profiles of the incoming axial speed, circumferential speed, azimuthal vorticity $(\eta = u_x - w_r)$, temperature, and reactant mass fraction. Thus, for every $0 \le r \le 1$,

$$\begin{aligned} w(0,r) &= w_0(r), \\ v(0,r) &= \omega v_0(r), \\ \eta(0,r) &= \eta_0(r), \\ T(0,r) &= 1 + \delta T_0(r), \\ Y(0,r) &= \delta Y_0(r), \end{aligned} \qquad (19)$$

where the maximum values of $w_0(r)$, $v_0(r)$ and $Y_0(r)$ are each taken to be unity. These given profiles are independent one of the other. Here, ω reflects the swirl ratio of the incoming flow, while δ is a measure of the amount of the incoming mass of reactant. The present study concentrates on lean mixtures where $0 \le \delta \ll 1$, and near-critical swirl ratios where $|\omega - \omega_1|/\omega_1 \ll 1$. As is typical of low-speed combustion, the limit $M_0 \ll 1$ is invoked as well. Note that the radial velocity component along the inlet is not prescribed. Instead, the inlet state is given the degree of freedom to develop a radial velocity, thus accommodating the upstream influence of disturbances that may develop downstream of the inlet. In the present study we also concentrate on the case where the incoming azimuthal vorticity is given by $\eta_0 = -w_{0r}$ or where there is a zero axial gradient of the radial velocity at the inlet, i.e., $u_x(0,r) = 0$. This assumption does not limit the present analysis and results. Analysis of the effects of general azimuthal vorticity profiles at the inlet is described in Rusak[35] and can be included in the current approach in a similar way but is beyond the scope of the present study. At the tube outlet $x = x_0$ we prescribe the radial velocity to be zero and the axial gradients are assumed to vanish, in accordance with an expected columnar flow state, i.e.,

$$\begin{aligned} u(x_0,r) &= w_x(x_0,r) = v_x(x_0,r) \\ &= T_x(x_0,r) = Y_x(x_0,r) = 0 \end{aligned} \qquad (20)$$

for every $0 \le r \le 1$.

Similar flow boundary conditions have been considered in the cold flow analysis of Wang and Rusak[28] and in the various numerical simulations of swirling flows in a tube. These conditions may also reflect the physical situation as reported in the cold flow experiments of Malkiel et al.[36] and Bruecker and Althaus.[37] They also formulate a basic problem which focuses on the interaction between the lean combustion state and the near-critical swirling flow.

The problem described by (9)-(16) with boundary conditions (17)-(20) for the solution of the velocity, temperature, pressure, and density fields constitutes a well defined mathematical problem. The solution of this nonlinear problem is a complicated task. It is also expected that under certain sets of inlet conditions the problem may have more than one steady-state solution.

The present paper explores the nonlinear interaction between the lean combustion resulting from the small amount of reactant entering the tube ($0 \le \delta << 1$), the low compressibility effects resulting from the low Mach number of the inlet flow ($0 \le M_0 << 1$), and the near-critical level of swirl, $|\omega - \omega_1| << 1$.

III. Small-disturbance approach

It is expected that the energy release associated with the combustion of the $O(\delta)$ amount of reactant entering the tube creates correspondingly small perturbations in temperature, pressure and density, as well as in the velocity components. Yet, previous studies on near-critical swirling flows show that in this range of swirl, velocity perturbations can be larger than those in the thermodynamic state.

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Accordingly, we postulate the asymptotic expansions

$$
\begin{aligned}
Y &= \delta Y_1 + \cdots, && (21)\\
T &= 1 + \delta T_1 + \delta_2 T_2 + \cdots, && (22)\\
p &= 1 + \gamma M_0^2 (p_0 + \epsilon_1 p_1 + \epsilon_2 p_2 + \cdots), && (23)\\
\rho &= 1 + \delta \rho_1 + \delta_2 \rho_2 + \cdots, && (24)\\
w &= w_0(r) + \epsilon_1 w_1 + \epsilon_2 w_2 + \cdots, && (25)\\
v &= \omega v_0 + \epsilon_1 v_1 + \epsilon_2 v_2 + \cdots, && (26)\\
u &= \epsilon_1 u_1 + \epsilon_2 u_2 + \cdots, && (27)\\
K &\equiv rv = K_0(r) + \epsilon_1 K_1 + \epsilon_2 K_2 + \cdots, && (28)
\end{aligned}
$$

where K denotes the circulation. It is assumed that $0 \leq M_0^2 \ll \delta \ll \epsilon_1 \ll 1$, specifically as the incoming flow swirl ratio ω approaches its critical level ω_1.

III.1. Leading-order pressure

In view of the above expansions, the radial and axial momentum equations, (11) and (13), yield at leading order,

$$p_{0_r} = \frac{\omega^2 v_0^2(r)}{r}, \quad p_{0_x} = 0, \tag{29}$$

whence the leading-order pressure is determined to be

$$p_0 = \omega^2 \int_0^r v_0^2(r')/r' dr'.$$

III.2. The basic flame structure

Similarly, the equation of state (9) and the species, energy and rate equations, (14) - (16), reduce at leading order to

$$\rho_1 + T_1 = 0, \tag{30}$$

$$w_0 Y_{1x} = \frac{1}{Pe}\left(\frac{1}{r}(rY_{1r})_r + Y_{1xx}\right) - \mathcal{W}, \tag{31}$$

$$w_0 T_{1x} = \frac{\mathcal{L}}{Pe}\left(\frac{1}{r}(rT_{1r})_r + T_{1xx}\right) + \beta\mathcal{W}, \tag{32}$$

$$\mathcal{W} = \mathcal{A} Y_1 \exp(-\theta/T).$$

Here, $T = 1 + \delta T_1$ is used. Typically, Pe is very large and effects of diffusion may be neglected everywhere except in the vicinity of the flame front. Boundary conditions for Y_1 and T_1 come from (17)-(19), and are

$$
\begin{aligned}
&Y_1(0,r) = Y_0(r),\ Y_{1x}(x_0,r) = 0 \\
&\qquad \text{for every } 0 \leq r \leq 1, \\
&Y_{1r}(x,0) = 0,\ Y_{1r}(x,1) = 0 \\
&\qquad \text{for every } 0 \leq x \leq x_0, \qquad (33)\\
&T_1(0,r) = T_0(r),\ T_{1x}(x_0,r) = 0 \\
&\qquad \text{for every } 0 \leq r \leq 1, \\
&T_{1r}(x,0) = 0,\ T_{1r}(x,1) = 0 \\
&\qquad \text{for every } 0 \leq x \leq x_0. \qquad (34)
\end{aligned}
$$

Equations (31) and (32), together with boundary conditions (33) and (34), constitute the standard, premixed flame-structure problem, and show that the reactant mass fraction perturbation Y_1 and the temperature perturbation T_1 can be computed for any given inlet axial flow $w_0(r)$. Moreover, the solution of the fields of Y_1, T_1, and $\rho_1 = -T_1$ is decoupled from the solution of the flow-speed perturbations. Of specific interest may be the case where we employ the approximation $T \sim 1$. Then the flame-structure problem becomes linear and can be solved by analytical methods. Another interesting situation is the case where the heat-release parameter is very large, $\theta \gg 1$. In this case the problem becomes in the leading order a semi-one dimensional problem and the flame structure can be analyzed by asymptotic methods.

III.3. Flame-induced flow perturbations at near-critical swirl

We now concentrate on the effects of reaction-generated density and temperature fields on the flow.

III.3.1. The stream functions

Turning first to the continuity equation (10), it reduces at leading order ϵ_1 to

$$\frac{1}{r}(ru_1)_r + w_{1x} = 0.$$

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The velocity perturbations u_1 and w_1 may now be written in terms of a stream function ψ_1 as

$$u_1 = -\frac{\psi_{1x}}{\sqrt{2y}}, \quad w_1 = \psi_{1y}. \tag{35}$$

where

$$y = r^2/2 \tag{36}$$

is a modify radial coordinate, varying in the interval $0 \leq y \leq 1/2$. We note that ψ_1 is known only to within an arbitrary constant. However, the first of the inlet boundary conditions (19) implies that $\psi_{1_y}(0,y) = 0$, so that ψ_1 is a constant at the inlet. Furthermore, the first of the symmetry conditions (17) on the tube axis implies that $\psi_{1_x}(x,0) = 0$, so that ψ_1 is a constant on the tube axis as well; in fact the same constant as that at inlet. We take this constant to be zero, so that

$$\psi_1(0,y) = \psi_1(x,0) = 0. \tag{37}$$

At the next order the continuity equation (10) reduces to

$$\epsilon_2 \left(u_{2r} + \frac{u_2}{r} + w_{2x} \right) + \delta w_0 \rho_{1x} = 0$$

and proceeding as above, we can define a second stream function ψ_2 via the relations

$$u_2 = -\frac{\psi_{2x}}{\sqrt{2y}}, \quad w_2 + \frac{\delta}{\epsilon_2} w_0 \rho_1 = \psi_{2y}. \tag{38}$$

By the same argument as above, ψ_2 can be taken to be zero on the axis, i.e.,

$$\psi_2(x,0) = 0. \tag{39}$$

At the inlet, however, the first of the boundary conditions (19) allows the second of the equations (38) to be written as

$$\psi_{2_y}(0,y) = \frac{\delta}{\epsilon_2} \rho_1(0,y) w_0(y), \tag{40}$$

which must be integrated subject to (39). Note that $\rho_1(0,y) = -T_1(0,y) = -T_0(y)$.

III.3.2. The circulations

We now turn to the circumferential momentum equation (12) and rewrite it in terms of the circulation K and the new radial coordinate y as

$$\sqrt{2y} u K_y + w K_x = 0. \tag{41}$$

At leading order ϵ_1 it yields

$$\sqrt{2y} u_1 K_{0_y} + w_0 K_{1_x} = 0,$$

and in view of the first of equations (35), can be re-expressed as

$$-\psi_{1_x} K_{0_y} + w_0 K_{1_x} = 0.$$

An integration, subject to the first of the boundary conditions (37), leads to

$$K_1 = \frac{K_{0y}}{w_0} \psi_1. \tag{42}$$

At order ϵ_2, (41) has the form

$$\sqrt{2y} u_2 K_{0_y} + w_0 K_{2_x} + \frac{\epsilon_1^2}{\epsilon_2} (\sqrt{2y} u_1 K_{1_y} + w_1 K_{1_x}) = 0.$$

Upon introducing the stream functions ψ_1 and ψ_2 via (35) and (38), and the circulation K_1 via (42), the above equation can be cast into

$$-\psi_{2_x} K_{0_y} + w_0 K_{2_x} - \frac{\epsilon_1^2}{\epsilon_2} \left(\frac{K_{0y}}{w_0} \right)_y \left(\frac{\psi_1^2}{2} \right)_x = 0.$$

Since K_2 and ψ_1 vanish at $x = 0$ (see the second of conditions (19) and the second of conditions(37)), the above equation integrates to yield

$$K_2 = \frac{K_{0y}}{w_0} [\psi_2 - \psi_2(0,y)] + \frac{\epsilon_1^2}{\epsilon_2} \left(\frac{K_{0y}}{w_0} \right)_y \frac{\psi_1^2}{2w_0}. \tag{43}$$

III.3.3. The eigenmode problem for critical swirl

The radial and axial momentum equations, (11) and (13), have already been used at order unity to determine the leading-order pressure

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p_0. We now write them each at the next two orders. Thus, (11) yields

$$\epsilon_1\left(w_0u_{1x}-\frac{2K_0}{r^3}K_1\right)+\epsilon_2\left(w_0u_{2x}-\frac{2K_0}{r^3}K_2\right)$$
$$+\epsilon_1^2\left(u_1u_{1r}+w_1u_{1x}-\frac{K_1^2}{r^3}\right)-\delta\frac{K_0^2}{r^3}\rho_1$$
$$=-\epsilon_1p_{1_r}-\epsilon_2p_{2_r}.$$

Upon substituting for K_1 from (42) and for K_2 from (43), the above equation may be rewritten as

$$\epsilon_1\left(w_0u_{1x}-\frac{2K_0}{r^3}\frac{K_{0y}}{w_0}\psi_1\right)$$
$$+\epsilon_2\left(w_0u_{2x}-\frac{2K_0}{r^3}\frac{K_{0y}}{w_0}[\psi_2-\psi_2(0,y)]\right)$$
$$+\epsilon_1^2\left[u_1u_{1r}\quad+w_1u_{1x}\right.$$
$$\left.-\frac{1}{r^3}\left\{\left(\frac{K_{0_y}}{w_0}\right)^2+\frac{K_0}{w_0}\left(\frac{K_{0y}}{w_0}\right)_y\right\}\psi_1^2\right]$$
$$-\delta\frac{K_0^2}{r^3}\rho_1=-\epsilon_1p_{1_r}-\epsilon_2p_{2_r}.$$

It is convenient to replace K_0 in favor of $\tilde{K}_0\equiv K_0/\omega=rv_0$, and let $\Omega=\omega^2$. Then the above equation may be expressed yet again as

$$\epsilon_1\left(w_0u_{1x}-\frac{2\tilde{K}_0}{r^3}\frac{\tilde{K}_{0y}}{w_0}\Omega\psi_1\right)$$
$$+\epsilon_2\left(w_0u_{2x}-\Omega\frac{2\tilde{K}_0}{r^3}\frac{\tilde{K}_{0y}}{w_0}[\psi_2-\psi_2(0,y)]\right)$$
$$\epsilon_1^2\left[u_1u_{1r}\quad+w_1u_{1x}\right.$$
$$\left.-\Omega\frac{1}{r^3}\left\{\left(\frac{\tilde{K}_{0_y}}{w_0}\right)^2+\frac{\tilde{K}_0}{w_0}\left(\frac{\tilde{K}_{0y}}{w_0}\right)_y\right\}\psi_1^2\right]$$
$$-\delta\Omega\frac{\tilde{K}_0^2}{r^3}\rho_1=-\epsilon_1p_{1_r}-\epsilon_2p_{2_r}. \qquad (44)$$

We now turn to the axial momentum equation (13) which reads

$$\epsilon_1\left(w_{0r}u_1+w_0w_{1x}\right)+\epsilon_2\left(w_{0r}u_2+w_0w_{2x}\right)$$
$$+\epsilon_1^2\left(u_1w_{1r}+w_1w_{1x}\right)=-\epsilon_1p_{1_x}-\epsilon_2p_{2_x},$$

and, upon a slight rearrangement, as

$$\epsilon_1\left(w_{0r}u_1+w_0w_{1x}\right)$$
$$+\epsilon_2\left[w_{0r}u_2+w_0\left\{w_2+\frac{\delta}{\epsilon_2}w_0\rho_1\right\}_x\right]$$
$$+\epsilon_1^2\left(u_1w_{1r}+w_1w_{1x}\right)-\delta w_0^2\rho_{1_x}$$
$$=-\epsilon_1p_{1_x}-\epsilon_2p_{2_x}. \qquad (45)$$

Elimination of pressure from (44) and (45) by cross-differentiation and subtraction leads to an equation for the azimuthal vorticity. At order ϵ_1, this equation reads

$$\left(w_0u_{1x}-\frac{2K_0}{r^3}\frac{K_{0y}}{w_0}\psi_1\right)_x-(w_{0r}u_1+w_0w_{1x})_r=0.$$

The use of (35), followed by some rearrangement, allows it to be rewritten as

$$\left[\psi_{1_{yy}}+\frac{\psi_{1_{xx}}}{2y}+\left\{\Omega\frac{\tilde{K}_0\tilde{K}_{0y}}{2y^2w_0^2}-\frac{w_{0_{yy}}}{w_0}\right\}\psi_1\right]_x=0.$$

An integration with respect to x is immediate. Upon using the third boundary condition in (19), which gives $\psi_{1_{xx}}(0,y)=0$, and the second condition in (37), which gives $\psi_1(0,y)=0$, we obtain the partial differential equation for the solution of ψ_1

$$\psi_{1yy}+\frac{\psi_{1xx}}{2y}+\left(\Omega\frac{\tilde{K}_0\tilde{K}_{0y}}{2y^2w_0^2}-\frac{w_{0yy}}{w_0}\right)\psi_1=0, \qquad (46)$$

with boundary conditions

$$\psi_1(x,0)=0,\quad \psi_1(x,1/2)=0$$
$$\text{for every}\quad 0\le x\le x_0, \qquad (47)$$
$$\psi_1(0,y)=0,\quad \psi_{1x}(x_0,y)=0$$
$$\text{for every}\quad 0\le y\le 1/2. \qquad (48)$$

This boundary-value problem possesses nontrivial solutions only for specific values of Ω, the eigenvalues. Wang and Rusak[28] defined the first eigenvalue $\Omega_1=\omega_1^2$ as the "critical level of swirl for a straight pipe of finite length x_0." The corresponding eigenfunction can be written as

$$\psi_1=\psi_{1c}(x,y)=\Phi(y)\sin\left(\frac{\pi x}{2x_0}\right) \qquad (49)$$

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where Φ is determined by

$$\Phi_{yy} + \left(\Omega_1 \frac{\tilde{K}_0 \tilde{K}_{0y}}{2y^2 w_0^2} - \frac{\pi^2/4x_0^2}{2y} - \frac{w_{0yy}}{w_0}\right)\Phi = 0,$$
$$\Phi(0) = 0, \quad \Phi(1/2) = 0. \tag{50}$$

The critical swirl Ω_1 is a transcritical bifurcation point of the equilibrium solutions of the reactionless equations, and has been discussed extensively by Wang and Rusak[26–28] in their study of the vortex breakdown phenomenon.

III.3.4. The near-critical behavior

We now return to equations (44) and (45) to determine the amplitude of the leading-order flame-induced flow perturbation for swirl ratios ω near the critical value ω_1. Let $\Omega = \Omega_1 + \Delta\Omega$ where $|\Delta\Omega|/\Omega_1 \ll 1$. With this prescription the radial momentum equation (44), upto the order kept, rearranges into

$$\begin{aligned}
&\epsilon_1 \left(w_0 u_{1cx} - \Omega_1 \frac{2\tilde{K}_0}{r^3}\frac{\tilde{K}_{0y}}{w_0}\psi_{1c}\right) \\
&+\epsilon_2 \left(w_0 u_{2x} - \Omega_1 \frac{2\tilde{K}_0}{r^3}\frac{\tilde{K}_{0y}}{w_0}[\psi_2 - \psi_2(0,y)]\right) \\
&\quad +\epsilon_1^2 \left[u_{1c}u_{1cr} + w_{1c}u_{1cx}\right. \\
&\quad \left. -\frac{\Omega_1}{r^3}\left\{\left(\frac{\tilde{K}_{0_y}}{w_0}\right)^2 + \frac{\tilde{K}_0}{w_0}\left(\frac{\tilde{K}_{0_y}}{w_0}\right)_y\right\}\psi_{1c}^2\right] \\
&\quad -\delta\Omega_1 \frac{\tilde{K}_0^2}{r^3}\rho_1 - \epsilon_1 \Delta\Omega_1 \frac{2\tilde{K}_0}{r^3}\frac{\tilde{K}_{0_y}}{w_0}\psi_{1c} \\
&\qquad = -\epsilon_1 p_{1_r} - \epsilon_2 p_{2_r}.
\end{aligned} \tag{51}$$

Once again, we cross-differentiate and eliminate the pressure between equations (45) and (51). Following some algebraic manipulations we arrive at the result

$$\begin{aligned}
&\epsilon_1 \left[\psi_{1cyy} + \frac{\psi_{1cxx}}{2y} + \left(\Omega_1 \frac{\tilde{K}_0\tilde{K}_{0y}}{2y^2 w_0^2} - \frac{w_{0yy}}{w_0}\right)\psi_{1c}\right]_x \\
&+\epsilon_2 \left[\psi_{2yy} + \frac{\psi_{2xx}}{2y} + \left(\Omega_1 \frac{\tilde{K}_0\tilde{K}_{0y}}{2y^2 w_0^2} - \frac{w_{0yy}}{w_0}\right)\psi_2\right]_x \\
&+\epsilon_1^2 \left[\frac{1}{w_0}\left\{\frac{\Omega_1 \tilde{K}_0 \tilde{K}_{0_y}}{2y^2 w_0^2} - \frac{w_{0_{yy}}}{w_0}\right\}_y\right. \\
&\left. +\frac{\Omega_1}{2y^2 w_0}\left\{\left(\frac{\tilde{K}_{0_y}}{w_0}\right)^2 + \frac{\tilde{K}_0}{w_0}\left(\frac{\tilde{K}_{0_y}}{w_0}\right)_y\right\}\right]\left(\frac{\psi_{1c}^2}{2}\right)_x \\
&+\delta\left[\frac{\Omega_1 \tilde{K}_0^2 \rho_1}{4y^2 w_0} - 2w_{0_y}\rho_1 - w_0 \rho_{1_y}\right]_x \\
&+\epsilon_1 \Delta\Omega_1 \frac{\tilde{K}_0 \tilde{K}_{0_y}}{2y^2 w_0^2}\psi_{1c_x} = 0.
\end{aligned} \tag{52}$$

As we have already seen, the leading term in the above equation vanishes. The remainder integrates with respect to x and yields

$$\begin{aligned}
&\epsilon_2 \left[\psi_{2yy} - \psi_{2yy}(0,y) + \frac{\psi_{2xx}}{2y}\right. \\
&\left. + \left(\Omega_1 \frac{\tilde{K}_0\tilde{K}_{0y}}{2y^2 w_0^2} - \frac{w_{0yy}}{w_0}\right)\{\psi_2 - \psi_2(0,y)\}\right] \\
&+\epsilon_1^2 \left[\frac{1}{w_0}\left\{\frac{\Omega_1 \tilde{K}_0 \tilde{K}_{0_y}}{2y^2 w_0^2} - \frac{w_{0_{yy}}}{w_0}\right\}_y\right. \\
&\left. +\frac{\Omega_1}{2y^2 w_0}\left\{\left(\frac{\tilde{K}_{0_y}}{w_0}\right)^2 + \frac{\tilde{K}_0}{w_0}\left(\frac{\tilde{K}_{0_y}}{w_0}\right)_y\right\}\right]\left(\frac{\psi_{1c}^2}{2}\right) \\
&+\delta\left[\frac{\Omega_1 \tilde{K}_0^2(\rho_1 - \rho_1(0,y))}{4y^2 w_0} - 2w_{0_y}(\rho_1 - \rho_1(0,y))\right. \\
&\qquad \left. -w_0(\rho_{1_y} - \rho_{1_y}(0,y))\right] \\
&\qquad +\epsilon_1 \Delta\Omega_1 \frac{\tilde{K}_0 \tilde{K}_{0_y}}{2y^2 w_0^2}\psi_{1c} = 0,
\end{aligned} \tag{53}$$

where we have used the boundary conditions

$$\psi_2(x,0) = 0, \quad \psi_2(x,1/2) = 0 \quad \text{for every } 0 \le x \le x_0, \tag{54}$$

$$\psi_{2xx}(0,y) = 0, \quad \psi_{2x}(x_0,y) = 0 \quad \text{for every } 0 \le y \le 1/2, \tag{55}$$

while $\psi_2(0,y)$ is given by (40).

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III.3.5. The solvability condition

We define

$$\tilde{\psi}_2 = \psi_2 - \psi_2(0,y). \tag{56}$$

Then, (53) can be rewritten as

$$\begin{aligned}
\epsilon_2 &\left[\tilde{\psi}_{2yy} + \frac{\tilde{\psi}_{2xx}}{2y} + \left(\Omega_1 \frac{\tilde{K}_0 \tilde{K}_{0y}}{2y^2 w_0^2} - \frac{w_{0yy}}{w_0}\right)\tilde{\psi}_2\right] \\
&= -\epsilon_1^2 \left[\frac{\Omega_1}{y w_0^{3/2}} \left(\frac{\tilde{K}_0 \tilde{K}_{0_y}}{y w_0^{3/2}}\right)_y \right. \\
&\qquad \left. -\frac{1}{w_0}\left(\frac{w_{0_{yy}}}{w_0}\right)_y\right]\left(\frac{\psi_{1c}^2}{2}\right) \\
&-\delta\left[\frac{\Omega_1 \tilde{K}_0^2(\rho_1 - \rho_1(0,y))}{4y^2 w_0} - 2w_{0_y}(\rho_1 - \rho_1(0,y))\right. \\
&\qquad \left. -w_0(\rho_{1_y} - \rho_{1_y}(0,y))\right] \\
&\qquad -\epsilon_1 \Delta\Omega_1 \frac{\tilde{K}_0 \tilde{K}_{0_y}}{2y^2 w_0^2}\psi_{1c}.
\end{aligned}$$

where we have also taken the opportunity to rearrange the $O(\epsilon_1^2)$ term above. The above equation is a nonhomogeneous PDE for $\tilde{\psi}_2$, subject to homogeneous boundary conditions for $\tilde{\psi}_2$ similar to those in (47) - (48). Since the corresponding homogeneous problem has the nontrivial solution ψ_1, solvability requires that the forcing term above be orthogonal to ψ_1. Upon multiplication of the above equation by ψ_{1c}, followed by an integration over the domain $(0 \le x \le x_0, 0 \le y \le 1/2)$ and use of the homogeneous conditions for $\tilde{\psi}_2$, we obtain

$$\epsilon_1^2\, M_1 - \epsilon_1 \Delta\Omega\, M_2 + \delta\, M_3 = 0. \tag{57}$$

Here,

$$M_1 = -\int_0^{x_0}\int_0^{1/2}\left[\Omega_1 \frac{1}{y w_0^{3/2}}\left(\frac{\tilde{K}_0 \tilde{K}_{0y}}{y w_0^{3/2}}\right)_y - \frac{1}{w_0}\left(\frac{w_{0yy}}{w_0}\right)_y\right]\frac{\psi_{1c}^3}{2}dydx = \frac{4x_0}{3\pi}N_1, \tag{58}$$

$$M_2 = \int_0^{x_0}\int_0^{1/2}\frac{\tilde{K}_0 \tilde{K}_{0y}}{2y^2 w_0^2}\psi_{1c}^2 dydx = \frac{1}{2}x_0 N_2, \tag{59}$$

$$\begin{aligned}
M_3 = \int_0^{x_0}\int_0^{1/2}&\left[2w_{0y}(\rho_1 - \rho_1(0,y)) \right. \\
&+ w_0(\rho_{1y} - \rho_{1_y}(0,y)) \\
&\left. -\Omega_1 \frac{\tilde{K}_0^2}{4y^2 w_0}(\rho_1 - \rho_1(0,y))\right]\psi_{1c}dydx
\end{aligned} \tag{60}$$

where

$$N_1 = -\int_0^{1/2}\left[\Omega_1 \frac{1}{y w_0^{3/2}}\left(\frac{\tilde{K}_0 \tilde{K}_{0y}}{y w_0^{3/2}}\right)_y - \frac{1}{w_0}\left(\frac{w_{0yy}}{w_0}\right)_y\right]\frac{\Phi^3(y)}{2}dy, \tag{61}$$

$$N_2 = \int_0^{1/2}\frac{\tilde{K}_0 \tilde{K}_{0y}}{2y^2 w_0^2}\Phi^2(y)dy. \tag{62}$$

Equation (57) is a quadratic equation for flow-disturbance amplitude ϵ_1. For relevant rotating base flows such as the Rankine vortex, the Burgers' vortex, or the "Q-vortex" it may be shown that N_1 and N_2 are positive. Also, with $\rho_1 = -T_1$ (equation (30)), where T_1 is found from the solution of the flame-structure equations (31) and (32), we can rewrite M_3 as

$$\begin{aligned}
M_3 = -\int_0^{x_0}\int_0^{1/2}&\left[2w_{0y}(T_1 - T_1(0,y)) \right. \\
&+ w_0(T_{1y} - T_{1y}(0,y)) \\
&\left. -\Omega_1 \frac{\tilde{K}_0^2}{4y^2 w_0}(T_1 - T_1(0,y))\right]\psi_{1c}dydx.
\end{aligned} \tag{63}$$

From the combustion problem (30-(32) we expect that T_1 increases with x along the tube ($T_1 \ge T_1(0,y)$), and decreases with r or y ($T_{1_y} < 0$). Also, for axial jet flows coming into the tube, $w_0 \ge 0$ and $w_{0y} \le 0$ for every $0 \le y \le 1/2$. It is expected, therefore, that in most relevant cases M_3 is also positive.

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IV. Results

Equation (57) implies that

$$\epsilon_1^2 = O(\epsilon_1 \Delta\Omega) = O(\delta). \tag{64}$$

Thus, the amplitude of the flow perturbation near critical swirl is much larger than the amplitude of the thermal perturbation due to reaction. Further, there exists a real solution for ϵ_1 if and only if

$$|\Delta\Omega| \geq 2\frac{\sqrt{\delta M_1 M_3}}{M_2}, \tag{65}$$

and the solution is given by

$$\epsilon_1(\delta, \Delta\Omega) = \frac{\Delta\Omega M_2 \pm \sqrt{(\Delta\Omega)^2 M_2^2 - 4\delta M_1 M_3}}{2M_1}. \tag{66}$$

For the branch of solutions with $\omega < \omega_1$ ($\Delta\Omega < 0$), the solution with the $(+)$ sign in (66) describes a flow state with a small-amplitude disturbance and a slight flow deceleration near the centerline whereas that with the $(-)$ sign in (66) corresponds to a flow state with a large-amplitude disturbance and a significant flow deceleration near the centerline. A schematic description of these flow states is shown in Figure 1.

When (65) is an equality, i.e., for

$$\begin{aligned} \omega_{1\delta a}^2 = \Omega_{1\delta a} &= \Omega_1 - 2\frac{\sqrt{\delta M_1 M_3}}{M_2} \\ \omega_{1\delta b}^2 = \Omega_{1\delta b} &= \Omega_1 + 2\frac{\sqrt{\delta M_1 M_3}}{M_2}, \end{aligned} \tag{67}$$

there exist special equilibrium states, which are saddle fold bifurcation points of solutions of the axisymmetric reactive flow problem; see the scheme in Fig. 2. This figure displays the axial velocity along the centerline at the tube outlet,

$$w(x_0, 0) = w_0(0) + \frac{\Delta\Omega M_2 \pm \sqrt{(\Delta\Omega)^2 M_2^2 - 4\delta M_1 M_3}}{2M_1}, \tag{68}$$

as function of ω near ω_1. This parameter is used to represent the nature of the flow states. It can be seen that the bifurcation around $\omega_{1\delta a}$ and $\omega_{1\delta b}$ is quadratic. The critical swirl $\omega_{1\delta a}$ may also be considered as an asymptotic estimate of the critical swirl of a reactive flow in a straight tube. We find that the first critical swirl decreases as a result of the reaction in the flow. However, unlike the case of a cold incompressible flow in a straight tube ($\delta = 0$) where the critical swirl corresponds to a transcritical bifurcation, (see also Benjamin[38] and Leibovich and Kribus[39]), here the critical swirl corresponds to a limit point where solutions fold.

As regards stability, we recall that for a cold flow in a straight tube, the critical swirl is also a point of exchange of stability (Wang and Rusak[26,27]). Specifically, the branch of the columnar flow solutions has a stable mode of disturbance when $\omega < \omega_1$ and is unstable when $\omega > \omega_1$.

We expect that the stability characteristics of the various branches of solutions may be directly inherited from the mechanism for a straight tube, specifically, when the the Mach number and the incoming reactant are sufficiently small. Also, we anticipate that the limit points found in the present analysis are points of exchange of stability (see Fig. 3 for a schematic representation of the expected behavior). The branch of solutions that is close to the columnar states is linearly stable to small axisymmetric disturbances when $\omega < \omega_{1\delta a}$, whereas the branch of large disturbance solutions is unstable.

It is also expected that, as the amount of incoming reactant δ increases $\omega_{1\delta a}$ decreases. Moreover, when, δ is increased above a certain limit value, $\delta > \delta_{limit} \sim w_0^2(0) M_1^2 / M_3$ the fold in the branch of solutions with $\omega < \omega_1$ disappears and large-amplitude perturbations may gradualy appear in the flow as the swirl ratio ω is increased.

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V. Conclusions

The dynamics of a lean-premixed combustion state in a straight tube with an underlying low Mach number near-critical axisymmetric swirling flow can be studied by asymptotic methods. A small-disturbance approach is developed to carefully explore the nonlinear interaction between the flow swirl and the heat transfer through the combustion. The analysis shows that in the leading order the reaction problem can be decoupled from the flow problem. However, the effect of the heat addition on the flow behavior is important and has a nonlinear character. We find that a lean reaction in the flow breaks the transcritical bifurcation of solutions of a cold flow in a straight tube into two branches of solutions. These branches fold at limit swirl levels near the critical swirl with a finite gap between the two branches. This means that no near-columnar axisymmetric state can exist in a finite range of incoming swirl around the critical level; the flow must develop large disturbances in this swirl range which are related to the vortex breakdown states and also strongly affect the combustion state. It also suggests that the critical swirl decreases with the increase of the reactant amount. Beyond this swirl range, two equilibrium states may exist under the same inlet/outlet conditions. One describes a near columnar flow and the other a large-amplitude disturbance. However, when the amount of the reactant is increased this special behavior uniformly changes into a branch of solutions with no fold. These describe a gradually increasing steady disturbance as the swirl is increased.

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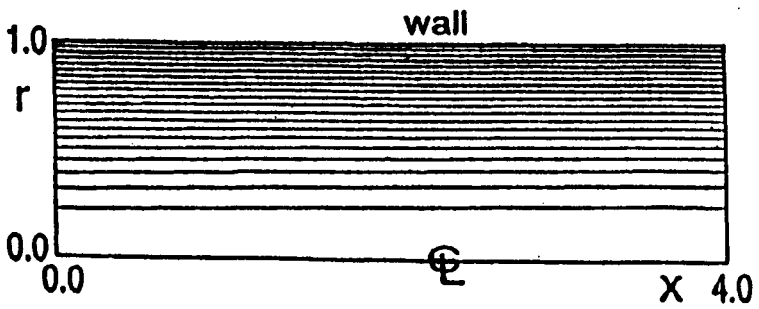

Figure 1a. A schematic description of streamline contours of a nearly columnar state (positive root of (57)).

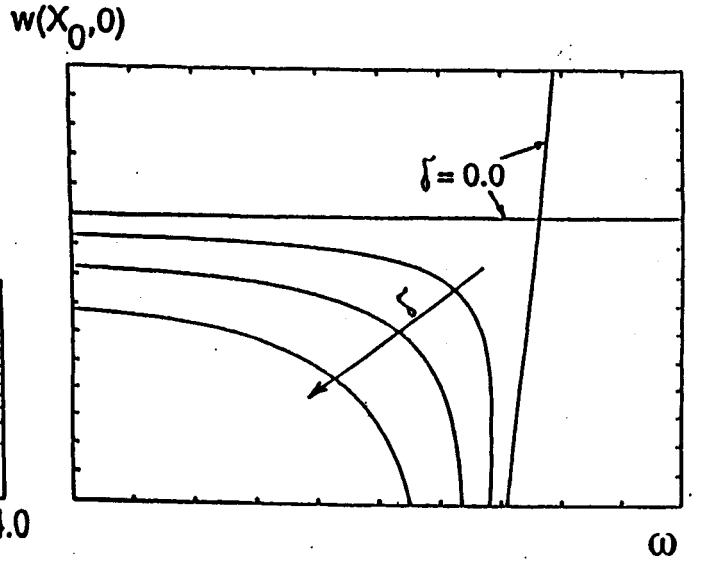

Figure 2. The bifurcation diagram of steady solutions of a reactive flow with swirl.

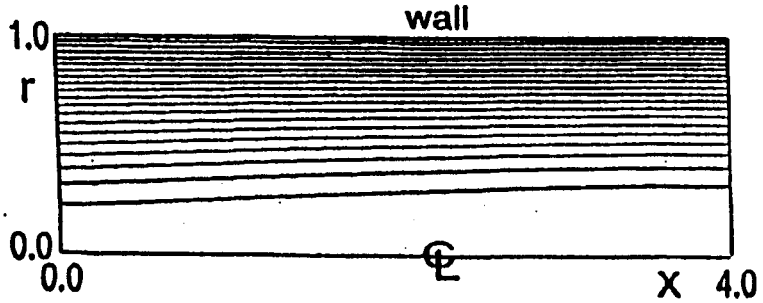

Figure 1b. A schematic description of streamline contours of a larger disturbance state (negative root of (57)).

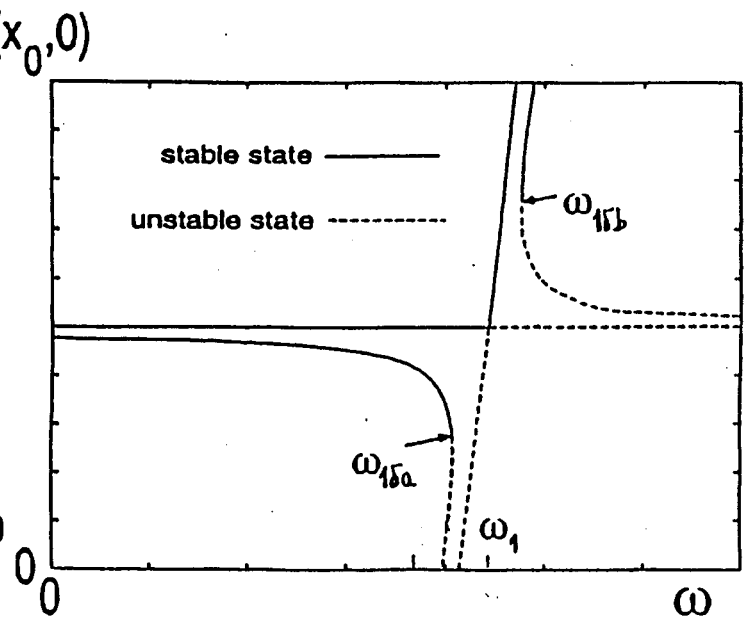

Figure 3. Stability characteristics of equilibrium states.